Chapter 3: N-gram Language Models - Exercises Solutions

Textbook: Speech and Language Processing (3rd ed. Draft) by Jurafsky, D. and Martin, J. H.

Here are the detailed, step-by-step solutions to the selected exercises from Chapter 3.

Exercise 3.1

Question: *Write out the equation for trigram probability estimation (modifying Eq. 3.11). Now write out all the non-zero trigram probabilities for the "I am Sam" corpus on page 4.*

Equation for Trigram Probability Estimation

The maximum likelihood estimate (MLE) for a standard bigram probability is conventionally given by normalizing the bigram count by the unigram history:

$$P(w_n|w_{n-1}) = \frac{C(w_{n-1}w_n)}{C(w_{n-1})}$$

Generalizing this strictly to a trigram model, we condition the probability of the current word on the **two preceding words**. We normalize the trigram sequence count by the count of its corresponding bigram prefix:

$$P(w_n|w_{n-2}w_{n-1}) = \frac{C(w_{n-2}w_{n-1}w_n)}{C(w_{n-2}w_{n-1})}$$

Non-zero Trigram Probabilities for the "I am Sam" Corpus

The original corpus established on page 4 consists of three sentences:

```
<s> I am Sam </s>
<s> Sam I am </s>
<s> I do not like green eggs and ham </s>
```

To properly compute trigram probabilities at the beginning of a sentence, we must assume two start pseudo-words (`<s> <s>`) to provide the necessary two-word context for the first actual string word.

The non-zero probabilities, grouped by their two-word histories, are formulated as follows:

From history `<s> <s>`: * $P(\text{I}|\text{<s>}, \text{<s>}) = \frac{C(\text{<s> <s> I})}{C(\text{<s> <s>})} = \frac{2}{3} \approx 0.67$ * $P(\text{Sam}|\text{<s>}, \text{<s>}) = \frac{C(\text{<s> <s> Sam})}{C(\text{<s> <s>})} = \frac{1}{3} \approx 0.33$

From history `<s> I`: * $P(\text{am}|\text{<s>}, \text{I}) = \frac{C(\text{<s> I am})}{C(\text{<s> I})} = \frac{1}{2} = 0.5$ * $P(\text{do}|\text{<s>}, \text{I}) = \frac{C(\text{<s> I do})}{C(\text{<s> I})} = \frac{1}{2} = 0.5$

From history `<s> Sam`: * $P(\text{I}|\text{<s>}, \text{Sam}) = \frac{C(\text{<s> Sam I})}{C(\text{<s> Sam})} = \frac{1}{1} = 1.0$

From history `I am`: * $P(\text{Sam}|\text{I}, \text{am}) = \frac{C(\text{I am Sam})}{C(\text{I am})} = \frac{1}{2} = 0.5$ * $P(\text{</s>}|\text{I}, \text{am}) = \frac{C(\text{I am </s>})}{C(\text{I am})} = \frac{1}{2} = 0.5$

From history `Sam I`: * $P(\text{am}|\text{Sam}, \text{I}) = \frac{C(\text{Sam I am})}{C(\text{Sam I})} = \frac{1}{1} = 1.0$

From history `am Sam`: * $P(\text{</s>}|\text{am}, \text{Sam}) = \frac{C(\text{am Sam </s>})}{C(\text{am Sam})} = \frac{1}{1} = 1.0$

Other unique deterministic sequences (all have a fractional probability of $\frac{1}{1} = 1.0$): * $P(\text{not}|\text{I}, \text{do}) = 1.0$ * $P(\text{like}|\text{do}, \text{not}) = 1.0$ * $P(\text{green}|\text{not}, \text{like}) = 1.0$ * $P(\text{eggs}|\text{like}, \text{green}) = 1.0$ * $P(\text{and}|\text{green}, \text{eggs}) = 1.0$ * $P(\text{ham}|\text{eggs}, \text{and}) = 1.0$ * $P(\text{</s>}|\text{and}, \text{ham}) = 1.0$

Exercise 3.2

Question: *Calculate the probability of the sentence "i want chinese food". Give two probabilities, one using Fig. 3.2 and the 'useful probabilities' just below it on page 6, and another using the add-1 smoothed table in Fig. 3.7. Assume the additional add-1 smoothed probabilities $P(i| < s >) = 0.19$ and $P(< /s > |food) = 0.40$.*

Calculating the Probability of "i want chinese food"

We estimate the global sentence probability by utilizing the primary bigram formulation defined by the Markov assumption constraints:

$$P(\text{<s> i want chinese food </s>}) = P(\text{i}|\text{<s>}) \times P(\text{want}|\text{i}) \times P(\text{chinese}|\text{want}) \times P(\text{food}|\text{chinese}) \times P(\text{</s>}|\text{food})$$

1. Unsmoothed Probability (Maximum Likelihood Estimation)

First, we pull the exact sequence derivations using the values from Figure 3.2 and the auxiliary 'useful probabilities': * $P(\text{i}|\text{<s>}) = 0.25$ * $P(\text{want}|\text{i}) = 0.33$ * $P(\text{chinese}|\text{want}) = 0.0065$ * $P(\text{food}|\text{chinese}) = 0.52$ * $P(\text{</s>}|\text{food}) = 0.68$

Calculation:

$$0.25 \times 0.33 \times 0.0065 \times 0.52 \times 0.68 \approx \mathbf{1.9 \times 10^{-4}}$$

2. Smoothed Probability (Add-1 Smoothing)

Conversely, utilizing the discrete Laplace add-one smoothed table detailed in Figure 3.7 alongside the custom constraints: * $P(\text{i}|\text{<s>}) = 0.19$ *(Explicitly* *declared in the exercise string)* * $P(\text{want}|\text{i}) = 0.21$ * $P(\text{chinese}|\text{want}) = 0.0029$ * $P(\text{food}|\text{chinese}) = 0.052$ * $P(\text{</s>}|\text{food}) = 0.40$ *(Explicitly declared in the exercise string)*

Calculation:

$$0.19 \times 0.21 \times 0.0029 \times 0.052 \times 0.40 \approx \mathbf{2.4 \times 10^{-6}}$$

Exercise 3.3

Question: *Which of the two probabilities you computed in the previous exercise is higher, unsmoothed or smoothed? Explain why.*

Comparison and Explanation

The **unsmoothed** (MLE) probability ($\approx 1.9 \times 10^{-4}$) inherently evaluates significantly higher than the processed **smoothed** probability ($\approx 2.4 \times 10^{-6}$).

Detailed Reasoning: Add-one (Laplace) smoothing works dynamically by assigning a mathematically uniform +1 to every potential structural n-gram matrix count mapped across the active vocabulary. By design, this fundamentally "steals" or discounts baseline probability mass directly from commonly observed n-grams (meaning the identical transitions mapped within the isolated testing string `"i want chinese food"`) and actively redistributes it outwardly to zero-count transitions that were never explicitly verified in the training layout. Because our specific string evaluates heavily verified transitions internally, their raw structural properties are steeply discounted by the generalized variance equation, resulting directly in the compounded probability metric shrinking greatly despite retaining similar scale properties.

Exercise 3.5

Question: *Suppose we didn't use the end-symbol* `</s>`*. Train an unsmoothed bigram grammar on the following training corpus without using the end-symbol* `</s>`*: a b; b b; b a; a a. Demonstrate that your bigram model does not assign a single probability distribution across all sentence lengths by showing that the sum of the probability of the four possible 2-word sentences over the alphabet {a,b} is 1.0, and the sum of the probability of all possible 3-word sentences over the alphabet {a,b} is also 1.0.*

Sentence Length Distribution without End-Symbols

Given the specific training corpus conditions:

```
<s> a b
<s> b b
<s> b a
<s> a a
```

We carefully format and evaluate our standard unsmoothed active bigram probabilities from their transitions: * $P(\text{a}|\text{<s>}) = 0.5$ and $P(\text{b}|\text{<s>}) = 0.5$ *(Each token functionally initiates exactly half the documented valid structural occurrences).* * $P(\text{a}|\text{a}) = 0.5$ and $P(\text{b}|\text{a}) = 0.5$ *(The character* `'a'` *is historically trailed exactly once by* `'a'` *and once by* `'b'`*).* * $P(\text{a}|\text{b}) = 0.5$ and $P(\text{b}|\text{b}) = 0.5$ *(The character* `'b'` *is historically trailed exactly once by* `'a'` *and once by* `'b'`*).*

Sum of Probability for 2-word Sentences

The explicit mathematically possible 2-word configurations built over our active vocabulary sequence space strictly amount to `aa`, `ab`, `ba`, and `bb`: * $P(\text{aa}) = P(\text{a}|\text{<s>}) \times P(\text{a}|\text{a}) = 0.5 \times 0.5 = 0.25$ * $P(\text{ab}) = P(\text{a}|\text{<s>}) \times P(\text{b}|\text{a}) = 0.5 \times 0.5 = 0.25$ * $P(\text{ba}) = P(\text{b}|\text{<s>}) \times P(\text{a}|\text{b}) = 0.5 \times 0.5 = 0.25$ * $P(\text{bb}) = P(\text{b}|\text{<s>}) \times P(\text{b}|\text{b}) = 0.5 \times 0.5 = 0.25$

Sum for Length 2: $0.25 + 0.25 + 0.25 + 0.25 = \mathbf{1.0}$

Sum of Probability for 3-word Sentences

Correspondingly, there exist exactly 8 valid structurally mapped 3-word sentences (`aaa`, `aab`, `aba`, `abb`, `baa`, `bab`, `bba`, `bbb`). Each uniquely mapped sequence breaks down to the core uniform factor: * $P(\text{aaa}) = P(\text{a}|\text{<s>}) \times P(\text{a}|\text{a}) \times P(\text{a}|\text{a}) = 0.5 \times 0.5 \times 0.5 = 0.125$ *(And uniformly consistently evaluated internally mapped equally for all exactly 8 variations).*

Sum for Length 3: $8 \times 0.125 = \mathbf{1.0}$

Conclusion

By structurally validating that internal probabilities for identical lengths uniquely evaluate explicitly bounding sums of 1.0 simultaneously across dimension variances (length 2 and length 3 uniformly evaluated independently sum equally to 1.0 globally), we successfully illustrate that the un-terminated structural language framework thoroughly fails to normalize naturally into a mathematically single probability distribution handling all arbitrary generalized length constraints dynamically.

Instead, lacking boundary terminations exclusively establishes infinite scaling bounds configuring explicitly nested distributions uniquely isolated internally relative to discrete length identifiers. Consequently, intentionally anchoring structural sequences properly utilizing token limits such as the defined </s> token becomes categorically mandatory locally to terminate sequence recursion bounds ensuring the structural domain strictly resolves normalized sequence possibilities bounding effectively to limits 1.0 internally distributed accurately scaling fully across dimensional boundaries explicitly spanning *any* dynamic variation sequences properly evaluating length mappings mathematically globally!